fuchs2026

Catherine Maximous

2026-10-17

Table of contents

Introduction . . . 2
Methods . . . 2
Learning Goals . . . 3
Power Analysis . . . 3
Planned Sample . . . 3
Materials . . . 4
Procedure . . . 4
Analysis Plan . . . 4
Differences from Original Study . . . 4
Methods Addendum (Post Data Collection) . . . 4
Results . . . 5
Data preparation . . . 5
Confirmatory analysis . . . 5
Exploratory analyses . . . 5
Discussion . . . 5
Summary of Replication Attempt . . . 5
Commentary . . . 5

[No abstract is needed.] Each replication project will have a straightforward, no frills report of the study and results. These reports will be publicly available as supplementary material for the aggregate report(s) of the project as a whole. Also, to maximize project integrity, the intro and methods will be written and critiqued in advance of data collection. Introductions can be just 1-2 paragraphs clarifying the main idea of the original study, the target finding for replication, and any other essential information. You will likely contact the authors of the original study, so this should be professional and forward facing. It will NOT have a literature review – that is in the original publication. You can write both the introduction and the methods in past tense.

Introduction

Recent research on language variability has demonstrated how different language environments shape language and cognitive processes. Specifically, Heritage bilinguals offer a powerful lens for understanding how becoming a dominant speaker in the second acquired language despite not having the first years of development in the language can provide insights into the plasticity of the language system. Therefore, understanding how language processing is shaped by the heritage bilingual experience and how that may differ from monolingual or second language learner is an important gap to address in the bilingualism literature.

I am proposing a new project proposal, where I would like to further explore heritage speakers' abilities to comprehend relative clauses across varying syntactic complexity. Therefore, in order to test this, I will be replicating Fuchs and colleagues (2026)'s paper where they compared heritage speakers to English-raised bilinguals who are late bilinguals and learned Spanish only in the classroom setting and Spanish-raised bilinguals who were raised in Spanish-speaking dominant environment. They tested participants' constraints when cognitive loading increased alongside syntactic complexity but also tested cognitive effort and syntactic complexity separately. For the purpose of this class, I will be replicating the experiment with heritage speakers in the picture-matching experiment which tested syntactic competence. The paper looked at three different types of sentences: Reduced Relative (RR), Subject Relative (SR), Object Relative (OR) in Spanish. RR tends to be the easiest condition as it follows a direct order of events for example "the bear painting the tiger." SR on the other hand is when the subject becomes the actor so instead it is "the bear that is painting the tiger." Finally, OR which is the most difficult syntactically as it starts with the noun where the sentence instead states "the bear that the tiger paints." In this sentence, it is typically harder for people as they often initially assume the bear is the one enacting the painting and not the tiger but then have to revise their interpretation once they hear the full sentence. The key findings of this experiment is that heritage speakers showed higher accuracy for Reduced Relatives than Subject Relatives ($= 0.87$, SE $= 0.22$, z $= 3.86$, p $< .001$) and significantly higher accuracy for Subject Relatives than Object Relatives ($= 5.07$, SE $= 0.17$, z $= 29.52$, p $< .001$), This means that heritage speakers tend to have difficulties in interpreting object relative clauses and show a relative-clause complexity effect.

Methods

To conduct this experiment, I will be implementing the experimental design from Fuchs and colleagues' (2026) design. Participants will hear a short Spanish phrase containing a relative clause and are instructed to select the picture that correctly matches the described event. Each trial will display two images depicting reversed agent-patient roles (e.g., a bear painting a tiger vs. a tiger painting a bear). The auditory stimuli will follow the structure of Article + Nouns + Relative Clause. As the task progresses, syntactic complexity will increase by using three types of relative clauses: Reduced Relatives, Subject Relative, and Object Relatives. All

materials for this experiment has been provided through the OSF repository: https://osf.io/et-prk/files/osfstorage?view_only=19f67921b4eb4ab58299685068ac0a16 along with analysis materials ensuring better replication of the experiment.

The paper recruited 31 heritage Spanish speakers, who were born and raised in the U.S. or immigrated by age 6 and reported speaking Spanish with one or both parents. They also report English to be their most dominant language but having acquired Spanish since birth. Ensuring that I am recruiting the correct population and including the same criteria as the researchers will be the most important and challenging part of the project. This is because it will restrict the availability of people I can recruit from SONA but will be the most fitting to what the study tested. Additionally, the organization and coding of the experiment on JavaScript will also be another learning challenge since participants completed 120 picture-matching trials so it is a substantive amount of images and audios to be coded.

Learning Goals

I am hoping through this project, I will learn how to build a task through jsPsych that is well structured. I have not had much experience with building new tasks especially in jsPsych however as I progress in building my second year project for the cognitive science program, I will need to design and build new tasks to fit my project aims. I also want to learn implementing similar analysis and methods in a published paper in order to implement some type of replication within a new project. This will help ensure that I am following up and building on these results and testing whether these effects exist before beginning a whole new project. I believe this project will be feasible as all the materials have been provided but I will need to ensure that I am organizing the task correctly and starting participant recruitment early to ensure I have enough participants for analysis. Additionally, this will be my first study looking at comprehension within bilinguals rather than production which will help inform my future projects towards assessing comprehension abilities and individual differences within heritage bilinguals.

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Planned sample size and/or termination rule, sampling frame, known demographics if any, preselection rules if any.

Materials

All materials - can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Procedure

Can quote directly from original article - just put the text in quotations and note that this was followed precisely. Or, quote directly and just point out exceptions to what was described in the original article.

Analysis Plan

Can also quote directly, though it is less often spelled out effectively for an analysis strategy section. The key is to report an analysis strategy that is as close to the original - data cleaning rules, data exclusion rules, covariates, etc. - as possible.

Clarify key analysis of interest here You can also pre-specify additional analyses you plan to do.

Differences from Original Study

Explicitly describe known differences in sample, setting, procedure, and analysis plan from original study. The goal, of course, is to minimize those differences, but differences will inevitably occur. Also, note whether such differences are anticipated to make a difference based on claims in the original article or subsequent published research on the conditions for obtaining the effect.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.